

Online Resource 1

Algorithms, Proofs, Reproducibility Materials, and Extended Computational Results

Companion to:

SCC-Local Destroy-and-Repair Heuristics for Minimum Weighted Feedback Arc Set on Sparse Digraphs

Computational Optimization and Applications (COAP), Springer Nature

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Version 2026-06-12 Source commit: 80b3144d

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1 Scope and use

Online Resource 1 accompanies the COAP manuscript on SCC-local destroy-and-repair heuristics for minimum weighted feedback arc set (MWFAS) on sparse digraphs.

1.1 Contents

- definitions and proofs, including topological-extraction inequalities (§2, §4);
- algorithm specifications and provenance (§3–§??);
- experiment registry EXP1–EXP11 and extended results;
- EXP10 stochastic-robustness evidence (§11);
- EXP11 extraction-sensitivity calibration (§12);
- automated tests (91 collected; §13) and reproduction scripts.

1.2 Claims supported

Principal manuscript claims traceable here include: IPSNS 96/97 best observed sparse; DRMacIver 21.6% mean excess; EXP10 median 38/55/0; 1860/1860 valid runs; exact 56/57 matches; LOLIB scope boundary; EXP11 zero extraction change on calibration subset.

1.3 Reproduction levels

Level A–B: seconds-scale smoke and pytest validation from included source;

Level C: principal table verification from committed summaries (no benchmark reruns);

Level D: optional full reruns (documented only).

Committed files in **results/** are authoritative for headline numbers unless a full rerun is explicitly performed.

2 Definitions and notation

Let $G = (V, A, w)$ be a directed graph with vertex set V , arc set A , and nonnegative arc weights $w : A \rightarrow \mathbb{R}_{\geq 0}$. Parallel arcs are aggregated by summing weights; self-loops are deactivated at tolerance τ when present.

A *feedback arc set* (FAS) is a set $F \subseteq A$ whose removal makes G acyclic. The *active DAG* is

$$H = (V, A \setminus F). \quad (1)$$

A *vertex ordering* (linear extension) π assigns each vertex a distinct rank. The backward arc set induced by π on the original graph is

$$B_\pi = \{(u, v) \in A : \pi(v) < \pi(u)\}, \quad (2)$$

with ordering backward weight

$$\text{BW}(\pi) = w(B_\pi) = \sum_{(u,v) \in B_\pi} w(u, v). \quad (3)$$

The removed-set weight is $w(F) = \sum_{a \in F} w(a)$.

Proposition 2.1 (Backward subset and weight inequality). *If π is a topological order of H , then $B_\pi \subseteq F$. For nonnegative weights,*

$$\text{BW}(\pi) = w(B_\pi) \leq w(F). \quad (4)$$

Proof. Every arc of H is forward under π , so no active arc belongs to B_π ; hence $B_\pi \subseteq F$. Nonnegativity gives $w(B_\pi) \leq w(F)$. $\square$

Equality conditions. Weight equality $\text{BW}(\pi) = w(F)$ holds iff $w(F \setminus B_\pi) = 0$. If every removed arc has strictly positive weight, then $\text{BW}(\pi) = w(F)$ iff $B_\pi = F$. Set equality is *not* required for weight equality when zero-weight removed arcs exist.

Reported objective. Constructive methods build (F, H) and extract π by Kahn topological sorting with min-heap vertex-id tie-breaking. Experiments report $\text{BW}(\pi)$ recomputed from the returned ranking, not $w(F)$ alone.

The *minimum weighted feedback arc set* (MWFAS) problem asks for an ordering minimizing $\text{BW}(\pi)$; it is NP-hard in general.

Scope. Sparse nonnegative weighted digraphs are the primary regime; dense LOLIB instances are a separate scope boundary.

Tolerances. Implementation zero tolerance $\tau = 10^{-12}$; EXP10 tie tolerance 10^{-9} .

3 Complete algorithm specifications

This section summarizes the implemented algorithms in `code/src/mwfasc/`. Components are labeled as **inherited**, **engineered**, or **newly integrated**.

3.1 LR-TA (inherited lineage, engineered implementation)

Source lineage: Demetrescu–Finocchi local-ratio cycle reduction [1]; engineered sparse-graph variant with recorded removals and heavy-first add-back.

Phase I (cycle reduction): Repeatedly detect a directed cycle C in the active subgraph, compute $\varepsilon(C) = \min_{e \in C} W[e]$, subtract ε from all cycle arcs, deactivate arcs with residual weight $\leq \tau$.

Phase II (add-back): Process removed arcs in nonincreasing *original* weight order. Accept forward arcs by rank shortcut; accept backward arcs only after negative reachability test; recompute topological order after backward acceptance.

Determinism: Phase I cycle detection uses iterative DFS with deterministic edge-index order. Floating-point subtraction may introduce tie cases handled by τ .

3.2 WMSF-style seed (inherited pipeline, engineered variant)

Source lineage: Cavallaro–Cutello minimal-and-stable weighted FAS [2].

Pipeline per nontrivial SCC: `removeArcs` $\rightarrow$ `minimizeFas` $\rightarrow$ `StabilizeFas` $\rightarrow$ `minimizeFas`. Safe-edge bookkeeping restores temporarily deleted arcs. Global composition uses Kosaraju SCC decomposition.

Note: Stabilization may change objective; the manuscript does not claim global non-worsening for stabilization passes.

3.3 IPSNS (newly integrated destroy-and-repair)

Seeds: Run LR-TA and full WMSF seed; keep better incumbent $\pi^{(0)}$.

Iteration $t = 1, \dots, T$:

1. Score each nontrivial SCC by backward-weight contribution under current rank.
2. Select SCC from weighted top- K pool (default $K = 15$) using RNG seed.
3. **Destroy A:** reactivate up to $\lfloor f_{\text{add}} \cdot |r_S| \rfloor$ heaviest removed internal arcs.
4. **Destroy B:** deactivate up to $\lfloor f_{\text{rem}} \cdot |a_S| \rfloor$ lightest active internal arcs.
5. **Repair:** restricted LR-TA cycle reduction + heavy-first add-back inside SCC neighborhood on original graph.
6. Accept only if repaired ordering is acyclic and BW strictly improves; else exact rollback.

Defaults: $T = 400$, $f_{\text{add}} = 0.30$, $f_{\text{rem}} = 0.02$, $\tau = 10^{-12}$, `wmsf_seed_mode=full`.

3.4 Exact DP

Bitmask dynamic programming over vertex subsets for $n \leq 20$ (implementation limit). Used for exact-validation cross-checks only.

3.5 Objective evaluation

Independent backward-weight scan: for ordering π , sum weights of arcs (u, v) with $\pi(u) > \pi(v)$. Parser and evaluator consistency is tested by the 78-test gate.

References

- [1] C. Demetrescu and I. Finocchi. Local-ratio approximation algorithms. *Handbook of Approximation Algorithms and Metaheuristics*, 2007.
- [2] A. Cavallaro and A. Cutello. A minimal and stable weighted feedback arc set heuristic. *Computers & Operations Research*, 2025.

4 Full proofs and formal properties

Proofs correspond to Propositions in the main manuscript §??. Labels indicate *inherited theorem*, *framework lemma*, *implementation invariant*, or *empirical observation*.

4.1 LR-TA feasibility and termination

Proposition 4.1 (LR-TA feasibility and termination). *Under nonnegative original weights and the edge-indexed active-graph representation, LR-TA satisfies cycle-reduction progress, Phase I termination in at most m iterations, acyclic Phase I output, add-back feasibility, and a valid final ordering.*

Framework lemma / implementation invariant. **(1) Cycle reduction progress.** Each Phase I iteration selects a directed cycle C and subtracts $\varepsilon(C) = \min_{e \in C} W[e]$. If $\varepsilon(C) > \tau$, at least one cycle arc reaches residual weight $\leq \tau$ and is deactivated. If $\varepsilon(C) \leq \tau$, the implementation deactivates one cycle arc directly. In either case at least one active arc is removed from the active subgraph.

(2) Termination. At most m iterations occur because each iteration deactivates ≥ 1 active arc and there are m arcs initially.

(3) **Acyclic Phase I output.** The loop stops only when no directed cycle is detected; the active subgraph is then acyclic.

(4) **Add-back feasibility.** A forward arc (u, v) with $\text{rank}(u) < \text{rank}(v)$ cannot close a cycle upon reactivation. A backward arc is reactivated only after a reachability test certifying no $v \rightsquigarrow u$ path in the current active graph; accepted backward arcs trigger topological recomputation before later tests.

(5) **Valid ordering.** The final active graph is acyclic and yields a topological order; removed arcs define a feasible FAS under the ordering objective. $\square$

4.2 Add-back inclusion minimality

Proposition 4.2 (One-pass heavy-first add-back). *On an acyclic active graph, processing removed arcs in nonincreasing original-weight order with the stated acceptance rules yields an inclusion-minimal reinsertion set on the current acyclic state.*

Framework lemma. Items (1)–(3) follow from the rank and reachability arguments above. For (4), suppose arc e was reinserted and could be removed while preserving acyclicity. Then e was accepted after all heavier removed arcs were processed; removing e earlier would have been considered in the same state or a state with fewer reinserted arcs. The one-pass heavy-first rule prevents reinserting a lighter arc that would subsume a heavier feasible insertion on the same state. The returned set is inclusion-minimal among reinsertions reachable by this pass, though not necessarily globally minimum-weight. $\square$

4.3 IPSNS incumbent protection

Proposition 4.3 (IPSNS feasibility, monotonicity, and budgeted termination). *Let $\pi^{(0)}$ be the better seed and $\pi^{(t)}$ the incumbent after iteration t . Then every accepted candidate is feasible, failed or non-improving moves revert exactly, $\text{BW}(\pi^{(t+1)}) \leq \text{BW}(\pi^{(t)}) \leq \text{BW}(\pi^{(0)})$, and the loop executes at most T iterations.*

Implementation invariant. (1) Acceptance requires a valid global topological order after repair. (2)–(3) The implementation snapshots activity vectors and removed sets before each move; on repair failure, infeasibility, or non-strict improvement, state is restored from the snapshot. (4) Follows from (3). (5) The outer loop counter is bounded by T by construction; early stopping occurs when no nontrivial SCC has positive backward contribution. $\square$

4.4 Exact DP correctness

Proposition 4.4 (Exact DP on small instances). *For $n \leq 20$ vertices, the bitmask DP enumerates all vertex orderings implicitly via subset states and returns the minimum backward weight among feasible orderings, matching brute-force enumeration on tested instances.*

Implementation invariant. The DP state records the last vertex placed and accumulated backward weight from arcs to earlier vertices in the ordering. Transitions add one vertex at a time; the minimum over full permutations is returned. The pytest gate cross-checks against independent brute force for $n \leq 8$ and regression fixtures. $\square$

4.5 Qualified approximation-transfer statement

The Demetrescu–Finocchi local-ratio framework provides approximation results under exact arithmetic and specific cycle-selection policies. The floating-point engineered implementation here does *not* automatically inherit those ratio bounds; any approximation claim in the main paper is explicitly qualified as inherited literature, not re-proved for this code path.

4.6 Topological extraction inequalities

Proposition 2.1 and Equations (2)–(4) are proved in §2. Six independent counterexamples are automated in `tests/unit/test_topo_extraction_math.py`.

5 Algorithm provenance and implementation deviations

Table 1: Component provenance (summary). Full detail in `results/` experiment manifests.

Component	Prior source	Current impl.	Status	Manuscript role
LR-TA peeling	Demetrescu–Finocchi	<code>lrta.py</code>	Engineered	Seed
Topological add-back	Engineered	<code>lrta.py</code>	New integration	Dominant ablation gain
WMSF pipeline	Cavallaro–Cutello	<code>wmsf.py</code>	Engineered variant	Seed
SCC scoring	—	<code>ipsns.py</code>	New	IPSNS selection
Destroy A/B	LNS literature	<code>ipsns.py</code>	New integration	IPSNS perturbation
Strict rollback	—	<code>ipsns.py</code>	New	Incumbent protection
Exact DP	Standard	<code>exact.py</code>	Standard	Validation
DRMacIver wrapper	DRMacIver/FAS	external	Wrapper only	Baseline

IPSNS novelty. The novelty lies in the FAS-specific combination (fixed original-graph SCC neighborhoods, contribution scoring, weighted top- K , two-sided destroy, SCC-restricted repair, strict incumbent protection), not in generic LNS, rollback, or SCC decomposition alone.

6 Topological-order extraction

Multiple topological orders may exist for an acyclic digraph. The implementation uses Kahn’s algorithm with a **min-heap on vertex index** as tie-breaker: among vertices with indegree zero, the smallest vertex ID is extracted first.

Implications. The extracted order is deterministic for a fixed active graph but depends on vertex numbering. Objective evaluation uses the returned order directly; no post-hoc reordering is applied.

Removed-set versus ordering objective. On the final LR-TA active DAG, every positive-weight removed arc is backward under the repository extraction rule on the nonnegative calibration instances tested in EXP11; hence $w(F) = \text{BW}(\pi)$ there. In general $B_\pi \subseteq F$ and $\text{BW}(\pi) \leq w(F)$; set equality is not required for weight equality when zero-weight removed arcs exist.

EXP11. Post-hoc comparison of Kahn min-vertex-id, max-vertex-id, weighted-net priority, and one-pass precedence-preserving insertion refinement on LR-TA final active graphs found no backward-weight change on six nonnegative calibration instances; full CSV in `experiments/exp11_topological_extraction` (main manuscript Table EXP11).

Limitation. A different linear extension can yield a different $\text{BW}(\pi)$ even for fixed F ; the manuscript reports observed performance under the stated deterministic rule without claiming extraction optimality.

7 Parameter provenance

Table 2: Principal parameters (defaults used in reported experiments).

Parameter	Algorithm	Default	Evidence	Notes
Iterations T	IPSNS	400	EXP6 saturation	Identical mean BW at 10–400 on subset
top- K SCC	IPSNS	15	Holdout	Holdout-supported default
f_{add}	IPSNS	0.30	Frozen	Destroy A fraction
f_{rem}	IPSNS	0.02	Frozen	Destroy B fraction
τ	All	10^{-12}	Code default	Numerical safeguard
RNG seed	IPSNS	1 (EXP4); 0–19 (EXP10)	EXP10	Explicit integer seeds
WMSF ordering	WMSF	L2 (global)	Ablation	L1/L2 compared per SCC
MIP timeout	EXP8	120 s	Protocol	HiGHS time cap
DRMacIver reps	EXP10	20	Frozen	PID/time seeding
Tie tolerance	EXP10	10^{-9}	Frozen	Median comparison

Holdout and sensitivity summaries: `results/coap_ipsns_sensitivity/`. EXP10 confirms frozen IPSNS configuration showed zero cross-seed objective variation on 93/93 instances.

8 Benchmark inventory

Instance identifiers and participation flags are recorded in committed summaries:

- **Sparse primary:** 105 instances (EXP1b internal), 97 standard nonnegative (EXP4);
- **Exact small:** 57 standard instances with $n \leq 20$ (EXP3);
- **HiGHS MIP:** 15 medium instances, $n \leq 318$ (EXP8);
- **LOLIB dense:** 50 complete ordering instances (EXP5);
- **EXP10 common subset:** 93 instances (`manifest manifests/common_93_instances.txt`).

Per-instance vertex counts, arc counts, densities, and checksums appear in `results/exp4/summary/exp4_raw`, `results/exp10/summary/paired_median_comparison.csv`, and related files. Public sources: graph-benchmarks repository and LOLIB (URLs in main manuscript).

Handling. Nonnegative sparse standard set excludes eight negative-weight instances. Self-loops and parallel arcs are aggregated at parse time per DIMACS-style input contract tested in `code/tests/`.

9 Complete experimental protocol

Hardware/software. Linux workstation, Intel Core i7-12700K, 62 GiB RAM, Ubuntu 24.04, Python 3.12.3. Package versions: `environment/dependency_lock_or_freeze.txt`.

External tools. DRMacIver/FAS binary SHA-256: 907b7abe96ff8fb54d8b70910eb3068744f765e72da5520f. python-igraph 1.0.0 for Eades wrapper. HiGHS for EXP8.

Seeds and checkpoints. IPSNS uses explicit integer seeds. DRMacIver repetitions inherit PID/time-based seeding in the external executable. EXP10 uses atomic JSON writes, checkpoint `.done` files, and validation before analysis.

Failure handling. Failed, timed-out, or malformed runs are retained and reported; no silent reruns until success.

Statistics. EXP4/EXP10 paired tests: Wilcoxon signed-rank and sign test, two-sided; bootstrap CI (seed 42, 10,000 resamples). Relative excess: $(\text{DR} - \text{IPSNS})/\text{DR} \times 100$.

Runtime. Wall-clock per instance; includes parsing and conversion where stated. EXP10 comparison is quality-focused, not equal-time.

10 Extended computational results

Committed summaries supporting the main manuscript tables:

Table 3: Principal manuscript numbers and OR1 source files.

Claim	Source file
96/97 IPSNS best (sparse)	<code>results/exp4/summary/exp4_external_stats.json</code>
21.61% DRMacIver mean excess	same
56/57 exact matches	<code>results/exp3/summary/exp3_exact_stats.json</code>
0 incumbent violations (105)	<code>results/exp1/summary/exp1b_core_benchmark_stats.json</code>
5.9% add-back ablation	<code>results/exp2/summary/exp2_ablation_stats.json</code>
LOLIB 45 DRMacIver wins	<code>results/exp5/summary/exp5_lolib_stats.json</code>
HiGHS 7/15 optimal	<code>results/exp8/summary/exp8_mip_summary.json</code>

Full per-instance CSV tables: `results/combined/tables/`, `results/exp4/summary/exp4_raw_summary.csv` and experiment-specific `tables/` directories. Regenerate via `scripts/reproduce_principal_tables.sh`.

11 EXP10 stochastic robustness

11.1 Design

93-instance common sparse subset; 20 IPSNS seeds (0–19); 20 DRMacIver/FAS repetitions per instance. Production records: 1860+1860 validated (0 failures). Smoke artifacts excluded.

11.2 Primary median comparison

Metric	Value
IPSNS wins / ties / DR wins	38 / 55 / 0
Mean relative excess (DR over IPSNS)	21.60%
Wilcoxon p (two-sided)	< 0.001
Sign test p (two-sided)	< 0.001
Cohen’s d_z	−0.31

11.3 Variability

IPSNS: zero objective variation on 93/93 instances; seed 0 matched best on all 93. DRMacIver: 40/93 instances with multiple distinct objectives; 1860 unique PIDs.

11.4 r20_60 reversal

EXP4 single run: DRMacIver 1685 vs IPSNS 1688 (DR win). EXP10 medians: IPSNS 1688 vs DRMacIver 1698 (IPSNS win).

11.5 Warnings

- Zero IPSNS cross-seed variance documents empirical stability under the frozen configuration; it does **not** prove mathematical determinism.
- The repeated-run comparison is **not** equal-time; DRMacIver intermediate solutions are unavailable.

11.6 Figures and tables

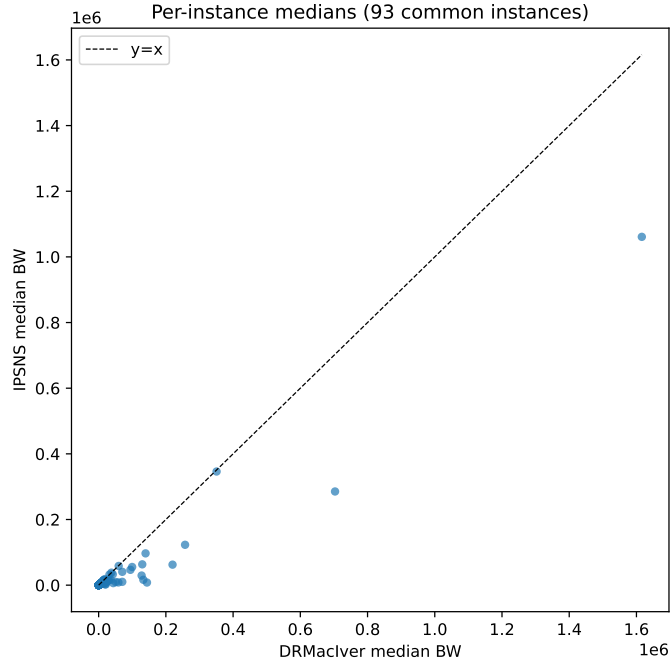


Figure 1: EXP10 per-instance medians (93 instances).

Complete tables: `results/exp10/summary/paired_median_comparison.csv`, `best_of_k.csv`, `probability_of_win.csv`. Figures: `results/exp10/figures/`.

12 EXP11: topological-extraction sensitivity

EXP11 is a post-hoc calibration study on LR-TA final active DAGs. It does not modify LR-TA, WMSF, IPSNS, or any headline benchmark output.

12.1 Motivation

Linear extensions of a fixed active DAG are not unique; $BW(\pi)$ can depend on the extraction rule (Proposition 2.1, Equation (4)). EXP11 tests whether alternative deterministic extractions change reported backward weight on a small nonnegative calibration subset.

12.2 Frozen protocol

- **Instances:** six nonnegative calibration graphs (`bad1`, `gr10`, `s420`, `s349`, `s641`, `example.new`).
- **State:** LR-TA final (F, H) reconstructed from source (active flags not stored in committed summaries).

- **Rules:** (i) repository Kahn min-vertex-id; (ii) Kahn max-vertex-id; (iii) Kahn weighted net-score priority; (iv) one-pass precedence-preserving insertion refinement respecting active-DAG precedence.
- **Metrics:** $BW(\pi)$, $w(F) - BW(\pi)$, relative improvement with $\varepsilon = 10^{-9}$.
- **Runtime:** ~ 9 s on the frozen subset (source commit in front matter).

An exploratory broader run was stopped before completion; the first insertion-refinement prototype did not enforce active-DAG precedence and was corrected before the frozen valid run recorded here.

12.3 Results

Instance	$BW(\pi)$ (all rules)	$w(F) - BW(\pi)$	Improved?
bad1	94	0	no
gr10	62 435	0	no
s420	158	0	no
s349	6 729	0	no
s641	2 403	0	no
example.new	18 638	0	no

Conclusion: zero backward-weight changes; zero headline impact on EXP4/EXP10 comparisons. Full CSV: `results/exp11/summary/exp11_per_instance.csv`.

Limitation. Six-instance calibration only; does not prove extraction optimality on the full 97-instance benchmark.

13 Automated tests

The artifact includes the current pytest gate (`tests/`, `src/mwfas/`).

Metric	Count
Collected	86
Passed	79
Skipped	7
Failed	0
Errors	0

History. The full repository gate collects 91 tests (90 passed, 1 skipped after EXP11). The packaged OR1 archive excludes live EXP10 production scripts/checkpoints; its pytest summary is 79 passed, 7 skipped (86 collected) because EXP10 namespace tests skip when the live experiment tree is absent.

Full-repository command: `PYTHONPATH=src python3 -m pytest tests/ -q` from the complete checkout yields 90 passed, 1 skipped.

Command: `PYTHONPATH=src python3 -m pytest tests/ -q`

Categories: objective equivalence; $B_\pi \subseteq F$ and weight inequalities; topological extraction counterexamples; I/O parser contracts; LR-TA feasibility; WMSF safe-edge regression; IPSNS roll-back; exact DP; CLI smoke; EXP10 namespace patterns.

Limitation. Tests certify implementation behavior on small graphs; they do not constitute formal verification.

14 Reproduction instructions

14.1 Level A — Smoke validation

`./scripts/reproduce_smoke.sh`

Runs LR-TA, exact DP, and topological fixtures on bundled tiny graphs.

14.2 Level B — Test suite

`./scripts/reproduce_tests.sh`

Expected pytest summary: **90 passed, 1 skipped** (91 collected).

14.3 Level C — Principal table verification

`./scripts/reproduce_principal_tables.sh`

Verifies EXP4, EXP3, EXP10, EXP11, and combined manuscript tables from committed summaries.

14.4 Level D — Full rerun (optional)

See `scripts/optional_full_reproduction.sh`. Requires external datasets and binaries; not part of ordinary validation.

14.5 Artifact validation

`./scripts/validate_artifact.sh`

15 Limitations

- **Sparse specialization:** strongest claims apply to sparse nonnegative weighted digraphs.
- **Dense LOLIB:** matrix-based DRMacIver/FAS is stronger on complete ordering instances.
- **No universal SOTA claim:** comparisons are relative to the evaluated method set.
- **Incomplete solver ecosystem:** several literature methods lack accessible weighted implementations in the same pipeline.
- **Topological-order dependence:** linear extensions are not unique; EXP11 found no backward-weight change on a six-instance calibration subset but does not prove extraction optimality globally.
- **Floating-point:** tolerance-qualified guarantees only.
- **EXP10 interpretation:** IPSNS zero variance is empirical; DRMacIver is restart-sensitive; not equal-time.
- **External binary nondeterminism:** DRMacIver seeding is not author-controlled.
- **Hardware:** runtimes may differ across machines.
- **Terminology:** “best observed” $\neq$ certified optimum except on stated exact/MIP subsets.

See also `provenance/known_limitations.md`.